

Koen Sakamoto

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EDUCATION

University of Utah

Anticipated May 2027

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Practices, Computer Security, Computer Organization, Computer Systems

TECHNICAL SKILLS

Languages: Java, Python, C#, SQL, C++, JavaScript/TypeScript, HTML/CSS

Frameworks & Libraries: Spring Boot, React, FastAPI, .NET, Selenium, Tailwind

Databases & Tools: MySQL, MongoDB, Docker, AWS, Git, Agile/Scrum, CI/CD

EXPERIENCE

Research Assistant

August 2025 – November

University of Utah

Salt Lake City, UT

- Contributed to research findings by benchmarking **15+ LLM Models** for vulnerability exposure.
- Engineered a custom **Model Context Protocol (MCP)** server and scalable testing pipeline to simulate tool interactions, enabling systematic detection and testing of backdoor triggers across multi-step LLM workflows.
- Developed interactive dashboards to visualize vulnerability metrics, test outcomes, and model performance trends

IT intern

April 2025 – July 2025

English Skills Learning Center

West Valley City, UT

- Architected automated asset management system using **database** design and **Python** scripts to inventory **300+ assets**, reducing manual overhead and preventing duplicate purchases.
- Resolved **40+ IT tickets** covering OS errors, hardware issues, and network problems, minimizing classroom and staff downtime and enabling uninterrupted daily operations.
- Created and maintained **10+ Standard Operating Procedures** for software installs, user permissions, and troubleshooting, empowering staff to self-resolve common issues without IT intervention.

PROJECTS

CrimsonBites – Campus Nutrition Tracking Platform | *FastAPI, React, MongoDB, Selenium*

- Built nutrition tracking platform helping **150+ students** make informed dietary choices by aggregating real-time menu data from 5+ campus dining halls with personalized dietary filtering.
- Engineered robust Selenium scraping pipeline with **cron scheduling** and **retry logic**, extracting **500+ daily** JavaScript-rendered menu items with **98% accuracy**.
- Created interactive React dashboards with **Recharts** displaying macronutrient trends, allowing students to track weekly averages and export nutrition reports.
- Deployed backend services on **AWS EC2** and offloaded static assets to **S3** for scalable and reliable hosting.
- Implemented authentication system with JWT tokens and Google OAuth 2.0, enabling secure session management

Rival Picks | *Spring Boot, MySQL, React Native*

- Created an app for friends to make friendly bets on anything from sports to personal challenges
- Implemented a real-time messaging system with **WebSocket + STOMP** and **Firestore Cloud Messaging**, enhancing team collaboration and engagement through instant notifications and live status updates.
- Built comprehensive security layer including BCrypt password hashing, rate limiting with **Bucket4j + Redis**, account lockout, and email verification flows, preventing brute force attacks and ensuring account integrity
- Managed database evolution with **Flyway** migrations, designing normalized schema with complex relationships and strategic indexing for query optimization

Snake Game | *C#, .NET, MySQL*

- Built a real-time multiplayer game server supporting **80+ concurrent players** using **TCP/IP sockets** and a custom protocol for collision detection and state sync.
- Developed a scalable **MySQL backend** with connection pooling and optimized queries to persist player data, leaderboards, and match statistics.
- Achieved smooth **60 FPS gameplay** with **multi-threading** for parallel input handling and real-time updates.